Cropland Recognition and Annotation

Team Sprint Review

27 September–11 October 2026

Matteo, Marc, Yassin, Wei and Matei

TU/e

ZERO_ HUNGER LAB

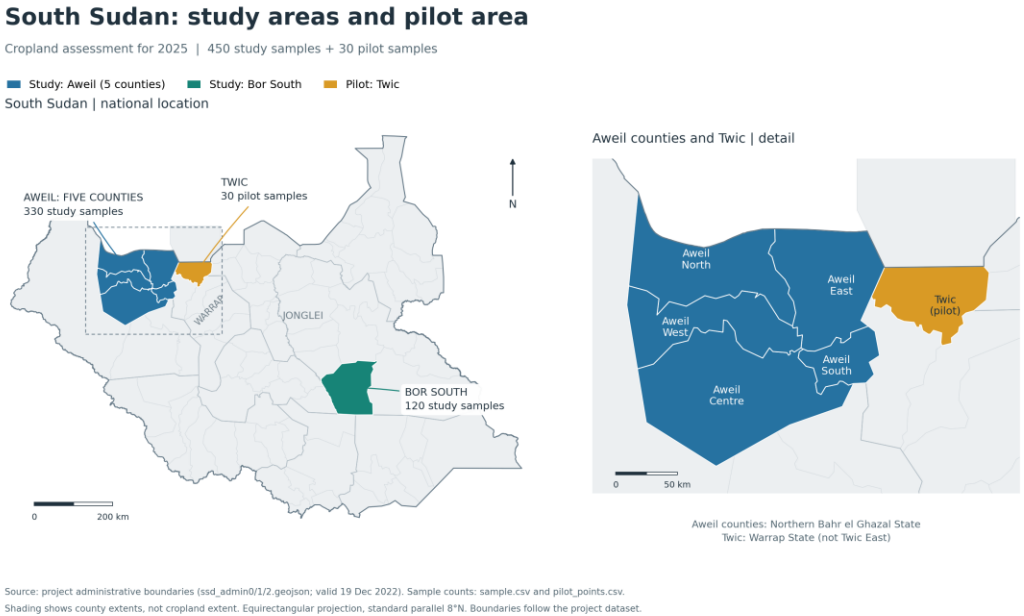
TILBURG UNIVERSITY

Research objectives and scope

2025 cropland

How much land was cropped or fallow in the two study areas?

Check public maps and develop a readable classification rule.

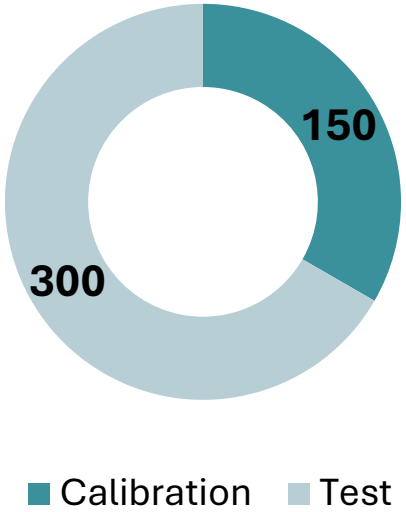


Sprint summary

28 Sep–4 Oct	5–11 Oct
Plan, guide and platform setup	210 m boxes and crop-share labels
All five members tested the pilot	Neutral charts and image overlay
Autosaving and platform fixes	900 formal judgements

Team progress in methodology, platform development and annotation

Methodology: independent labels



450 locations

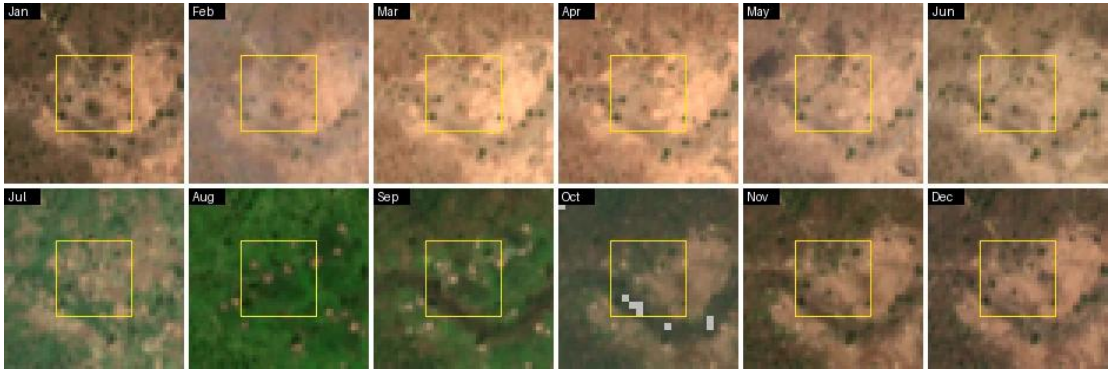
330 Aweil + 120 Bor South

2 independent labellers per point

180 assigned judgements each

The test split stays separate from model development.

Recognition principles



Monthly images: bare soil, growth and harvest timing

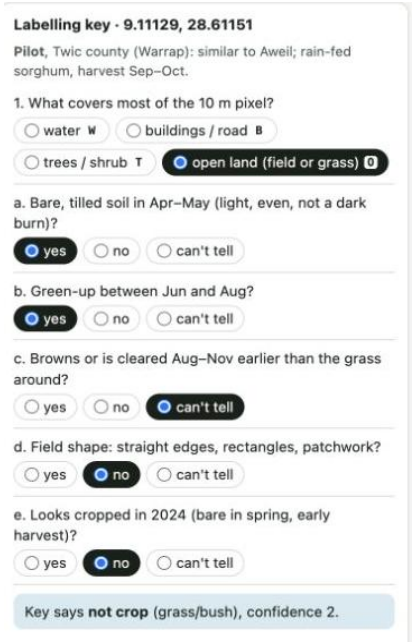


Field geometry

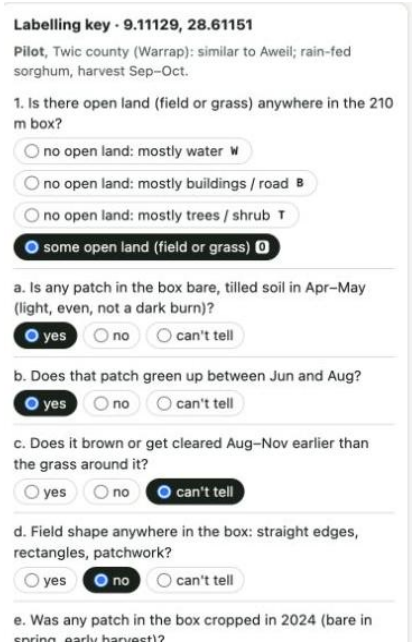
2024 helps identify fallow. Greenness alone cannot confirm cropping.

Methodological changes

Early page



Updated page

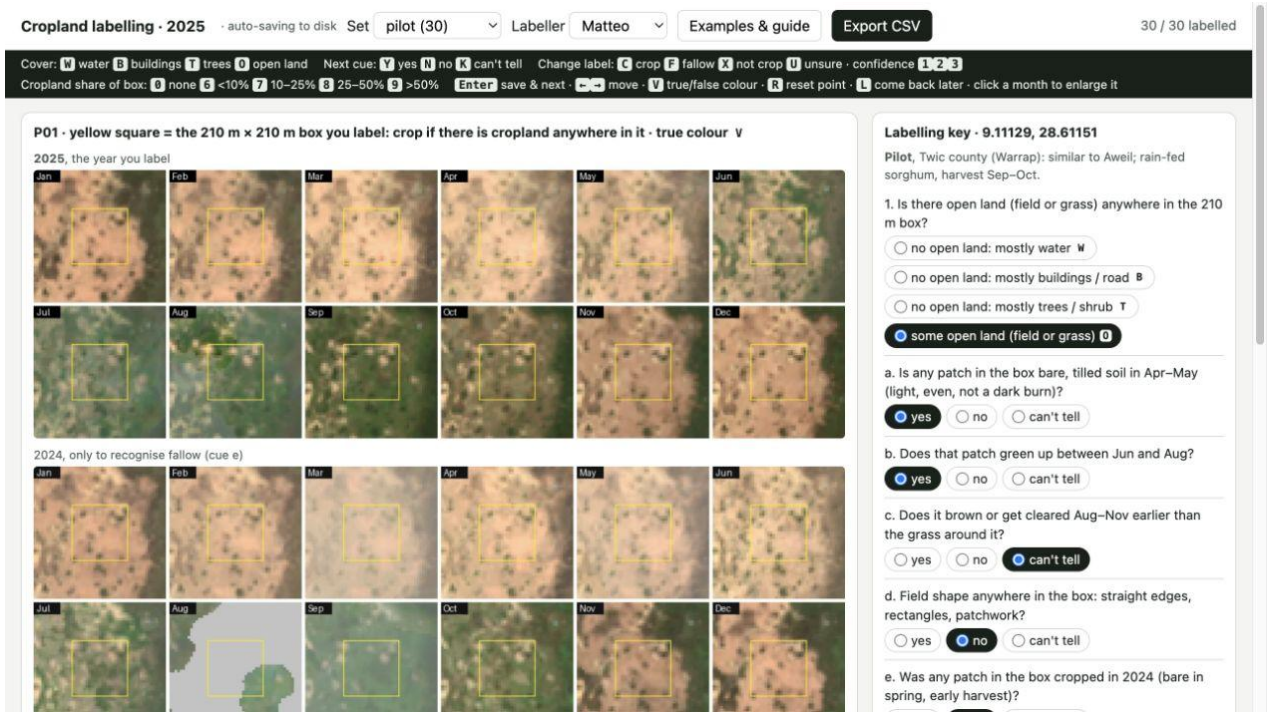


10 m pixel to 210 m box

The larger box is easier to interpret.

Crop-share categories separate field presence from estimated area.

Annotation platform



Monthly evidence

Guided questions

Autosave and shortcuts

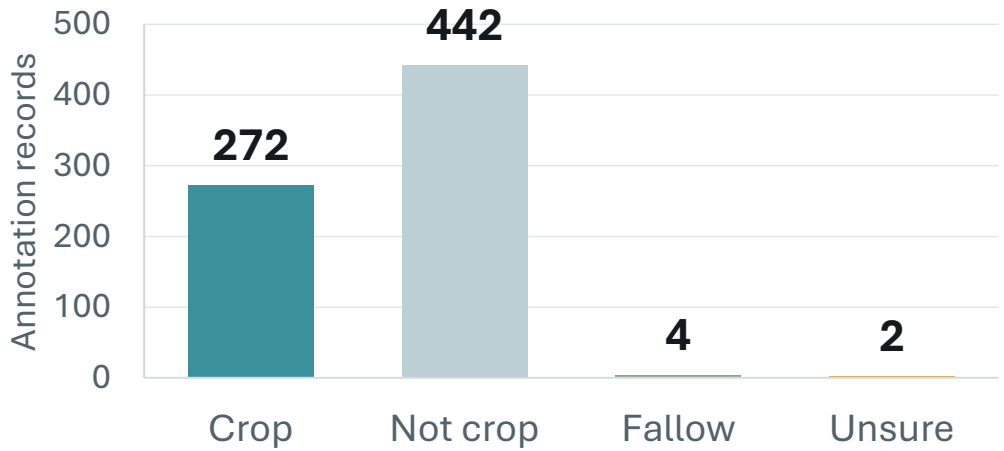
Page improvement: image overlay



Month zoom connects seasonal signals with field geometry.

Current annotation results

720 available judgements across 450 locations



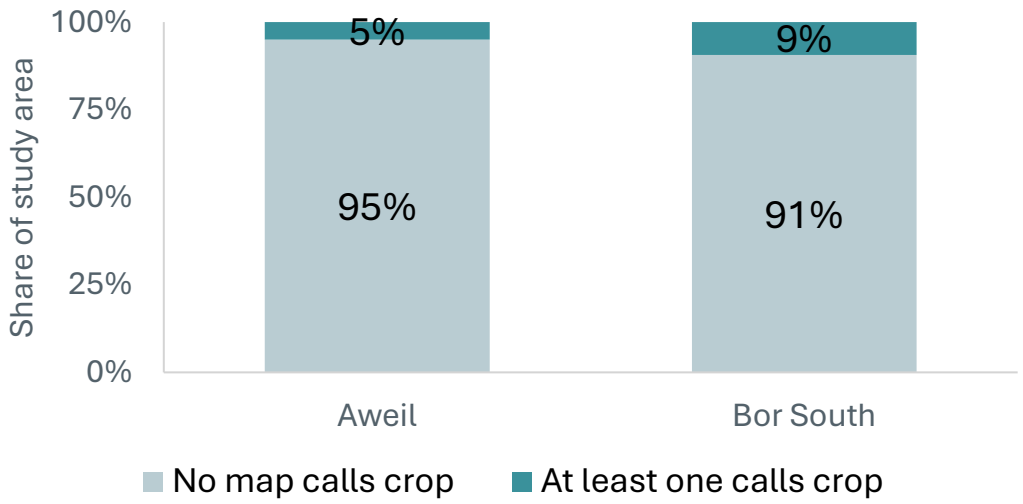
86.7%

exact agreement

234 of 270 paired locations
36 disagreements

Shared files, 11 Oct 2026. Counts describe labels, not cropland area.

Map EDA supports stratified sampling



Seven public maps

Few places receive crop votes.

We oversample those places, then weight results back to area.

Map agreement guides sampling. It is not ground truth.

Stakeholder relevance

“might not capture agricultural land fully”

ZOA on the supplied data

Our response

Check crop maps in Aweil and Bor South.

Report area and uncertainty separately.

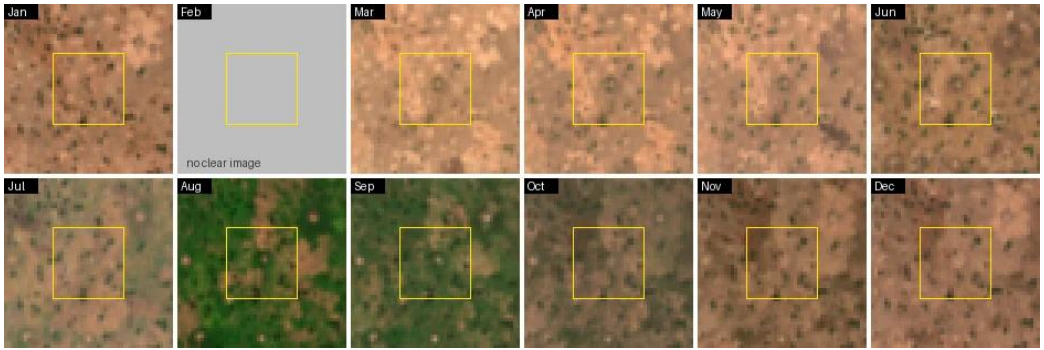
Keep the model interpretable.

Stakeholder outputs: next stage

Output	Planned approach
Cropland area	Box shares and sampling weights
Map quality	Score agreed labels in the test split
Readable model	Shallow decision tree, depth 2–4
Flood exposure	Cropland and flood map overlay

Current evidence: prepared inputs and annotations.
Model scores and final area estimates are still pending.

SLE: bias and uncertainty



Cloud gaps and image dates matter

Reference examples have no verified crop label.

Independent labels

Hide peer decisions and map votes.

Honest uncertainty

Keep “unsure” and disagreements. Validate each area separately.

Team lessons and next milestones

What we learned	What follows
Small units are hard to label	Use box labels and crop shares
Charts can guide the answer	Show neutral evidence
Shared files need checking	Confirm the remaining file
Test data needs protection	Freeze the rule, then score

Planned: rule freeze 12 Oct, results 15 Oct
Poster 21 Oct, report 25 Oct

Team reflection and discussion

Working well

All five members tested the workflow.

Team changes improved both the method and the page.

Shared files make progress traceable.

Points to resolve

Confirm the missing formal submission.

Agree how to explain label disagreement.

Finish model evaluation before making accuracy claims.